MILITARY STANDARD

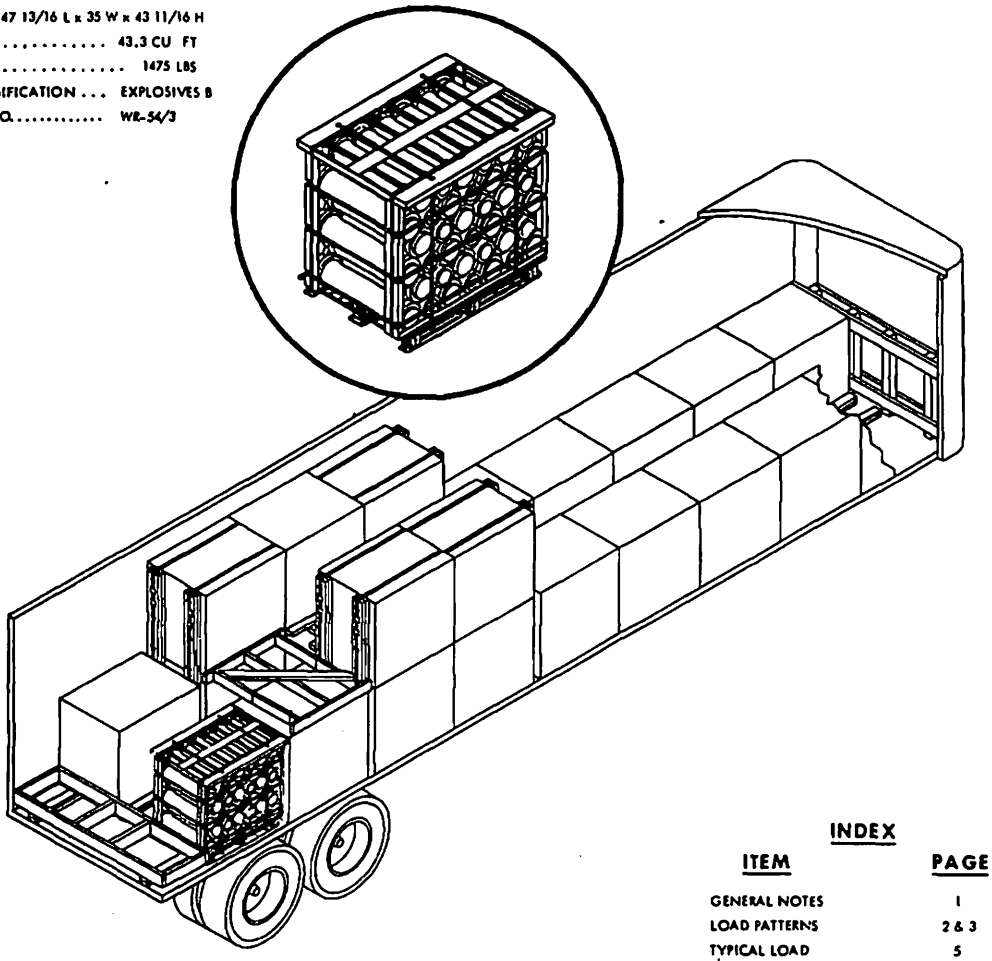
TRUCKLOADING OF HAZARDOUS MATERIALS

CHARGE, PROPELLING 5"/38
IN CARTRIDGE TANK MK 15
FLEET ISSUE UNIT LOAD

MIL-STD-1320-84A (NAVY)
26 MAY 1976
SUPERSEDING
MIL-STD-1320-84 (NAVY)
15 MAY 1974

UNIT LOAD DATA

UNIT LOAD DRAWING WR-54/3
DIMENSIONS ... 47 13/16 L x 35 W x 43 11/16 H
CUBE ..... 43.3 CU FT
WEIGHT (APPROX) ..... 1475 LBS
DOT HAZARD CLASSIFICATION ... EXPLOSIVES B
UNIT LOAD DWG NO. WR-54/3



NOTES:
1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS-REFERENCE TO ASSOCIATED PALLETIZING, CONTAINER-LOADING AND CARLOADING MILITARY STANDARDS. REFER TO INDEX TO STANDARDS, MIL-HDBK-236 (NAVY).

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FSC 8140

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GENERAL NOTES:

1. THIS DOCUMENT PROVIDES DETAILED INSTRUCTIONS FOR TRUCKLOADING 5"/38 PROPELLING CHARGES IN CARTRIDGE TANK MK 15 WHEN THE TANKS ARE PALLETIZED IN ACCORDANCE WITH FLEET ISSUE UNIT LOAD WR-54/3.
2. THE PROCEDURE AND PRACTICES DESCRIBED HEREIN ARE INTENDED FOR 40, 42, 44 AND 45 FT TRAILERS WITH BOTH NAILABLE AND NON-NAILABLE FLOORS. TRAILERS SHALL HAVE THEIR AXLES LOCATED IN THE "WESTERN" POSITION (AT THE EXTREME REAR OF THE TRAILER).
3. LENGTH OF THE STRUTS, PIECE 5 IN DETAIL A, SHALL BE SUFFICIENT TO CAUSE THE BLOCKING ASSEMBLY TO FILL THE VOID BETWEEN THE LADING AND THE TRAILER DOORS WHEN THE DOORS ARE CLOSED.

WARNING

DO NOT USE TRAILERS WITH ROLL UP DOORS.

4. TO PREVENT LONGITUDINAL MOVEMENT IN THE SECOND LAYER, THE UNIT LOADS SHALL BE UNITIZED (SEE UNITIZING DETAIL PAGE 7) WHERE THE PATTERN CHANGES FROM ONE LAYER HIGH TO TWO LAYERS HIGH. STIFFENERS SHALL FACE THE SINGLE LAYER SIDE. WHENEVER THERE IS A SINGLE STACKED UNIT LOAD (SEE LOAD PATTERN FOR 20 UNIT LOADS IN TABLE II) AND THE STACKED LOADS ARE NOT BACKED UP BY ANOTHER STACKED LOAD, THE STIFFENERS SHALL BE USED ON BOTH ENDS OF THE UNITIZED LOADS.
5. WHENEVER THERE ARE UNIT LOADS TWO HIGH ON ONE SIDE OF THE TRAILER AND ONE HIGH ON THE OTHER SIDE, SWAY BRACE ASSEMBLY, DETAIL B, SHALL BE USED AS SHOWN ON PAGE 5. THE LENGTH OF THE STRUTS, PIECE 12, SHALL BE SUFFICIENT TO CAUSE THE SWAY BRACE ASSEMBLY TO BEAR AGAINST THE UNIT LOAD AND THE TRAILER SIDE WALL. TWIST THE ASSEMBLY TO THE UNIT LOAD BENEATH (4 PLACES) USING 16 GAUGE SOFT ANNEALED IRON WIRE.
6. WHEN TRAILERS HAVE NON-NAILABLE FLOORS, SUBSTITUTE SWAY BRACE, DETAIL E, FOR SLEEPERS AND INSTALL BETWEEN BOTTOM LAYER PALLET AS PRESCRIBED IN NOTE 7 BELOW.
7. THE SWAY BRACE, DETAIL E, IS ASSEMBLED IN PLACE. INSERT SUPPORTS, PIECE 7, AGAINST INBOARD SIDE OF PALLET VERTICAL SUPPORTS. POSITION STRINGERS, PIECE 8, AGAINST PALLET VERTICAL SUPPORT AND NAIL IN PLACE WITH THREE 10d NAILS. WHEN USING SWAY BRACE IN LIEU OF SLEEPERS (PIECE 9), INSTALL BETWEEN THE LOWER LAYER PALLETS IN THE SAME MANNER.
8. THE MAXIMUM GROSS WEIGHT OF THE TRACTOR-TRAILER AND THE ALLOWABLE AXLE WEIGHTS ARE THE RESPONSIBILITY OF THE CARRIER. THE CARRIER WILL ADVISE THE SHIPPER OF THESE LIMITATIONS AND THE SHIPPER SHALL LOAD THE TRAILER IN SUCH A MANNER THAT THE TRACTOR-TRAILER WILL NOT EXCEED THESE LIMITATIONS.
9. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED, ATTACH SHIPPING DOCUMENTS TO INSIDE OF TRAILER IN AN ACCESSIBLE AREA, CLOSE AND SEAL TRAILER DOORS AND ATTACH "EXPLOSIVES B" PLACARD TO BOTH SIDES, FRONT AND BACK OF TRAILER.
10. APPLICABLE MATERIAL SPECIFICATIONS: DUNNAGE LUMBER, MM-L-751; NAILS, FF-N-105, TYPE II STYLE 10, COMMON BRIGHT; STRAPPING, QQ-S-781, TYPE I CLASS A.
11. FOR GENERAL TRUCKLOADING PROCEDURES REFER TO THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320 (NAVY).

FTL AND LTL 40, 42, 44 & 45 FT TRAILERS (VAN)

WARNING

THIS STANDARD PROVIDES FOR INCREASED LOADINGS WHICH PRODUCE HEAVIER GROSS TANDEM AXLE WEIGHTS AND GROSS VEHICLE WEIGHTS. THESE INCREASED WEIGHTS ARE NOT AUTHORIZED IN ALL 50 STATES. BEFORE LOADING MORE THAN 30 UNIT LOADS IN A TRAILER, DETERMINE THAT THE LIMITATIONS OF THE NOTES IN TABLES II, III & IV ARE MET

1. TABLES II, III & IV PROVIDE THE CORRECT LOAD PATTERNS FOR THE LENGTH OF THE TRAILER BEING USED AND THE NUMBER OF UNIT LOADS BEING LOADED.

NOTE

LIGHT WEIGHT TRACTOR-TRAILER - TARE WEIGHT OF TRAILER TANDEM AXLES 6,800 LBS MAXIMUM AND THE DRIVE TANDEM AXLES 10,500 LBS MAXIMUM.

STANDARD WEIGHT TRACTOR-TRAILERS - TARE WEIGHT OF TRAILER TANDEM AXLES 7,600 LBS MAXIMUM AND THE DRIVE TANDEM AXLES 11,500 LBS MAXIMUM.

2. TO DETERMINE THE CORRECT LOAD PATTERN TO USE SELECT THE CORRECT TABLE FOR THE LENGTH OF TRAILER BEING LOADED. OPPOSITE THE NUMBER OF UNIT LOADS TO BE LOADED, FIND THE CORRECT LOAD PATTERN. MAKE CERTAIN THAT THE QUANTITY BEING SHIPPED IS COMPATIBLE WITH THE WEIGHT OF THE TRAILER BEING LOADED AND THE WEIGHT LAWS OF THE STATES THROUGH WHICH THE SHIPMENT IS ROUTED. BLOCK AND BRACE USING THE PRINCIPLES OF THE TRUCKLOADS SHOWN ON PAGES 5 AND 6.
3. THE TRUCKLOAD SHOWN ON PAGE 5 DOES NOT LIMIT SHIPMENT TO THE NUMBER OF UNIT LOADS SHOWN. IT IS PRESENTED TO SHOW THE BASIC PRINCIPLES OF LOADING ONLY. THE LOAD PATTERN TABLES SHOW THE CORRECT NUMBER OF UNIT LOADS FOR THE LENGTH AND WEIGHT OF TRAILER BEING LOADED AND THE WEIGHT RESTRICTIONS OF THE ROUTING.
4. WHEN LOADING TRAILERS WITH A SECOND LAYER, FOLLOW THE PROCEDURES ON PAGE 5, INCREASING OR DECREASING THE NUMBER OF UNIT LOADS IN THE SECOND LAYER AS REQUIRED BY THE LOAD PATTERN.
5. WHEN LOADING A FULL SINGLE LAYER, BLOCK AT REAR OF TRAILER AS SHOWN ON PAGE 5.
6. WHEN SHIPPING LESS THAN THE AMOUNTS SHOWN IN TABLE II, FOLLOW THE PROCEDURES SHOWN ON PAGE 6. DO NOT USE TRAILERS WITH NON-NAILABLE FLOORS.

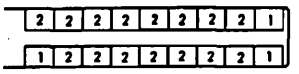
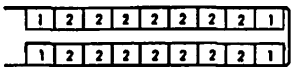
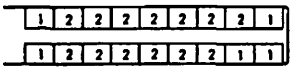
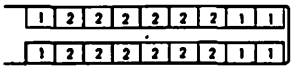
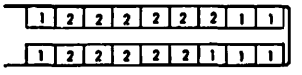
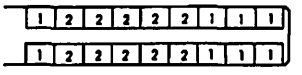
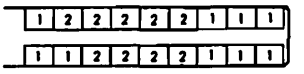
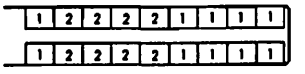
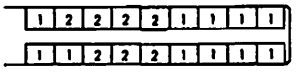
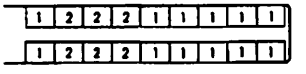
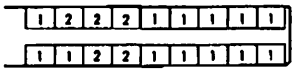
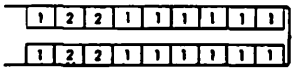
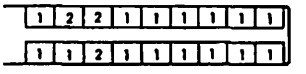
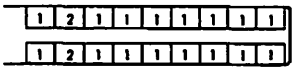
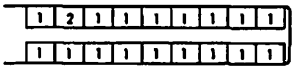
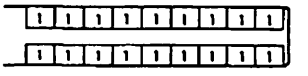
TABLE I
TRUCKLOAD WEIGHTS

NO. OF UNIT LOADS	TRUCKLOAD APPROX WT (LBS)
33	48,675
32	47,200
31	45,725
30	44,250
29	42,775
28	41,300
27	39,825
26	38,350

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TABLE II

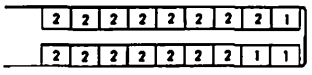
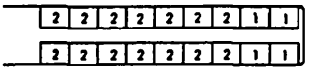
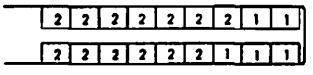
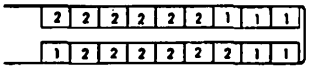
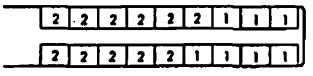
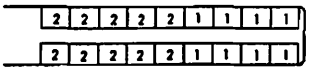
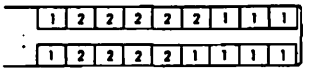
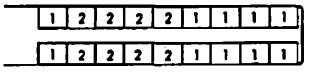
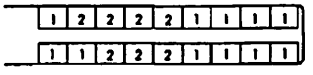
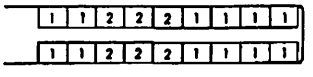
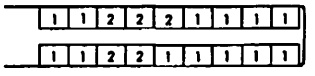
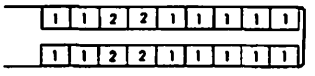
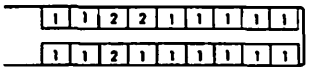
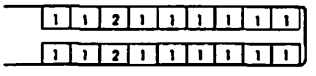
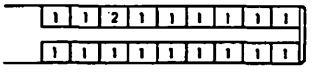
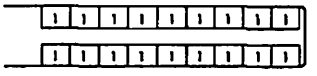
40 FT TRAILER

NO. OF UNIT LOADS	LOAD PATTERN → FWD
33****	
32***	
31**	
30*	
29*	
28*	
27*	
26*	
25*	
24*	
23*	
22*	
21*	
20*	
19*	
18*	

NOTE: LOAD PLAN SHOWS TRAILER FLOOR PLAN. NUMBERS IN BLOCKS INDICATE NUMBER OF UNIT LOADS IN A LAYER.

TABLE III

42 FT TRAILER

NO. OF UNIT LOADS	LOAD PATTERN → FWD
33****	
32***	
31**	
30*	
29*	
28*	
27*	
26*	
25*	
24*	
23*	
22*	
21*	
20*	
19*	
18*	

NOTE: LOAD PLAN SHOWS TRAILER FLOOR PLAN. NUMBERS IN BLOCKS INDICATE NUMBER OF UNIT LOADS IN A LAYER.

**** AUTHORIZED WHEN ROUTING IS THROUGH STATES PERMITTING 34,000 LBS OR MORE ON THE TANDEM AXLES AND A GROSS VEHICLE WEIGHT OF 80,000 LBS. ONLY LIGHT WEIGHT TRACTOR-TRAILERS SHALL BE USED.

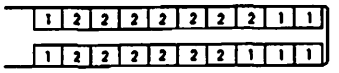
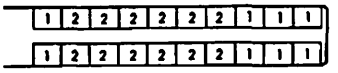
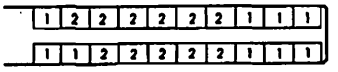
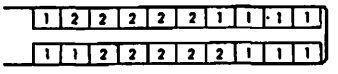
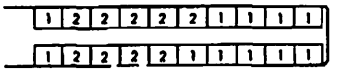
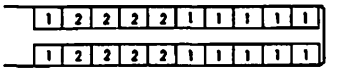
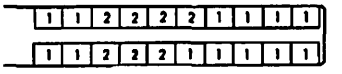
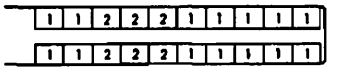
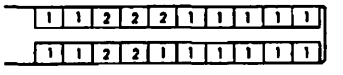
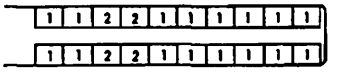
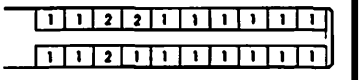
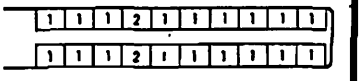
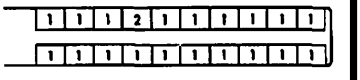
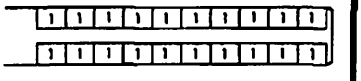
*** AUTHORIZED WHEN ROUTING IS THROUGH STATES PERMITTING 34,000 LBS OR MORE ON THE TANDEM AXLES AND A GROSS VEHICLE WEIGHT OF 80,000 LBS. BOTH STANDARD AND LIGHT WEIGHT TRACTOR-TRAILERS MAY BE USED.

** AUTHORIZED FOR LIGHT WEIGHT TRACTOR-TRAILERS ONLY WHEN ROUTING IS THROUGH STATES LIMITING TANDEM AXLES TO 32,000 LBS AND GROSS VEHICLE WEIGHTS OF 73,280 LBS. AUTHORIZED FOR BOTH STANDARD AND LIGHT WEIGHT TRACTOR-TRAILERS WHEN ROUTING IS THROUGH STATES PERMITTING 34,000 LBS ON TANDEM AXLES AND GROSS VEHICLE WEIGHTS OF 80,000 LBS.

* AUTHORIZED WHEN ROUTING IS THROUGH STATES PERMITTING 32,000 LBS ON THE TANDEM AXLES AND A GROSS VEHICLE WEIGHT OF 73,280 LBS. BOTH STANDARD AND LIGHT WEIGHT TRACTOR-TRAILERS MAY BE USED.

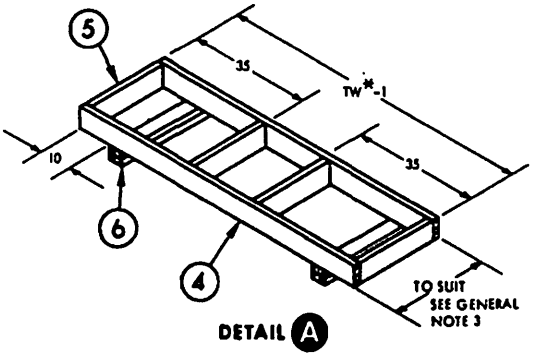
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TABLE IV
44 & 45 FT TRAILERS

NO. OF UNIT LOADS	LOAD PATTERN → FWD
33****	
32***	
31**	
30*	
29*	
28*	
27*	
26*	
25*	
24*	
23*	
22*	
21*	
20*	
****	
***	
**	
*	

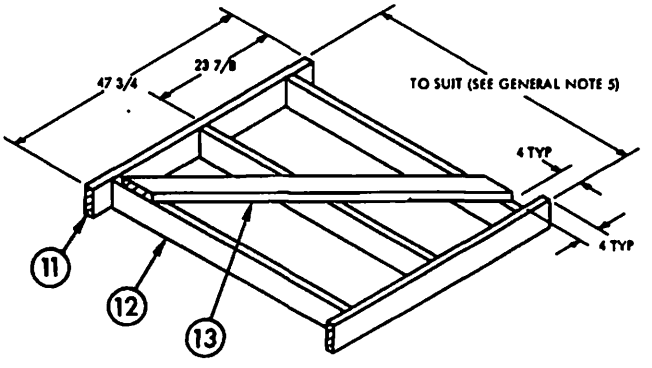
SEE PAGE 3

NOTE: LOAD PLAN SHOWS TRAILER FLOOR PLAN. NUMBERS IN BLOCKS INDICATE NUMBER OF UNIT LOADS IN A LAYER.

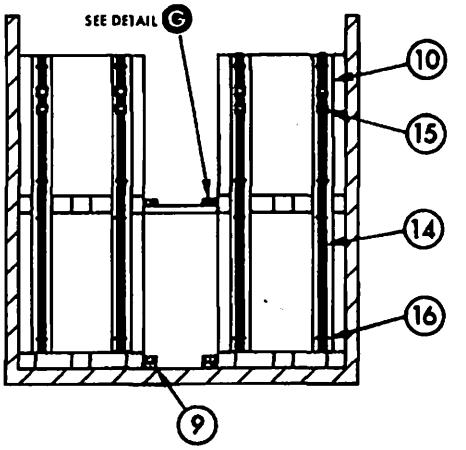


DETAIL A
REAR BLOCKING ASSEMBLY
1 REQUIRED

* TW = TRAILER WIDTH

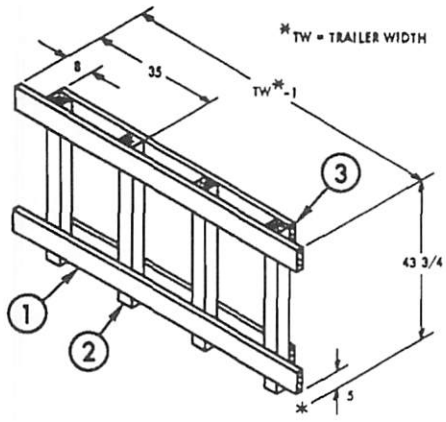


DETAIL B
SWAY BRACE ASSEMBLY
1 REQUIRED



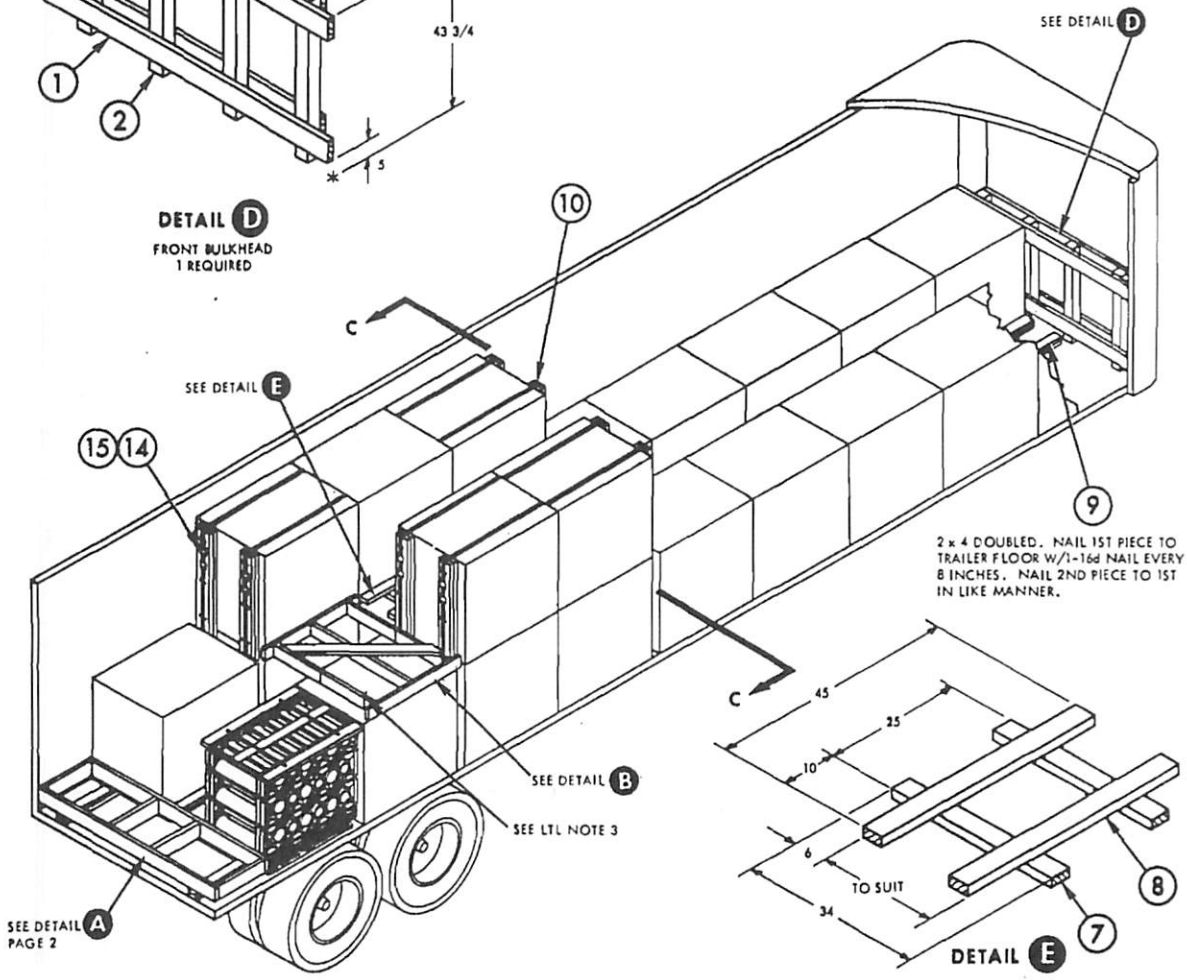
SECTION C-C

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DETAIL D
FRONT BULKHEAD
1 REQUIRED

NOTE
FOR TRAILERS WITH NON-NAILABLE FLOOR, SEE GENERAL NOTE 6.



DETAIL E
SWAY BRACE
2 REQUIRED
SEE GENERAL NOTE 7

*TW = TRAILER WIDTH

PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NUMBER NAILS	SIZE NAILS
16	STAPLE	FOR 1 1/4 STRAP	32	-	-	-
15	SEAL	FOR 1 1/4 STRAP	16	-	-	-
14	STRAP	1 1/4 x .035 x 24 FT	8	-	-	-
13	DIAGONAL	2 x 4 TO SUIT	1	12	2 PER JOINT	12d
12	STRUT	2 x 6 x TO SUIT	3	SEE 11.	-	-
11	LONGITUDINAL	2 x 6 x 47 3/4	2	12	3 PER JOINT	16d
10	STIFFENER	2 x 6 x 82 1/4	16	DBLD & LAMINATED	5	10d
9	SLEEPER	2 x 4 x 45	36	SEE FIELD NOTE		

LIST OF MATERIALS AND NAILING DATA

PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NUMBER NAILS	SIZE NAILS
8	STRINGER	2 x 4 x 45	4	7	3 PER JOINT	12d
7	SUPPORT	2 x 4 x 34	4	SEE 8	-	-
6	SUPPORT	2 x 4 TO SUIT DOUBLED	4	4 & 6	2 PER JOINT	10d
5	STRUT	2 x 6 TO SUIT	4	SEE 4	-	-
4	CROSSMEMBER	2 x 6 x TW*-1	2	5	3 PER JOINT	16d
3	FWD HORIZONTAL	2 x 6 x TO SUIT	2	2	3 PER JOINT	12d
2	VERTICAL	4 x 4 x 43 3/4	4	SEE 1 & 3	-	-
1	AFT HORIZONTAL	2 x 6 x TW*-1	2	2	3 PER JOINT	12d

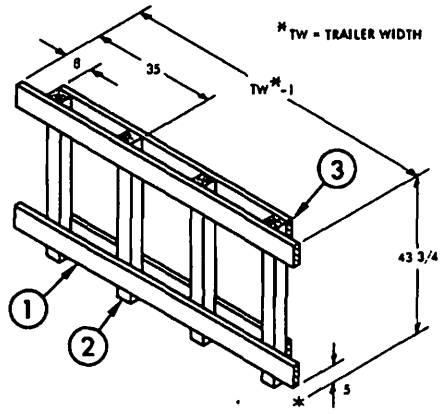
LIST OF MATERIALS AND NAILING DATA

TYPICAL TRUCKLOAD WITH PARTIAL SECOND LAYER

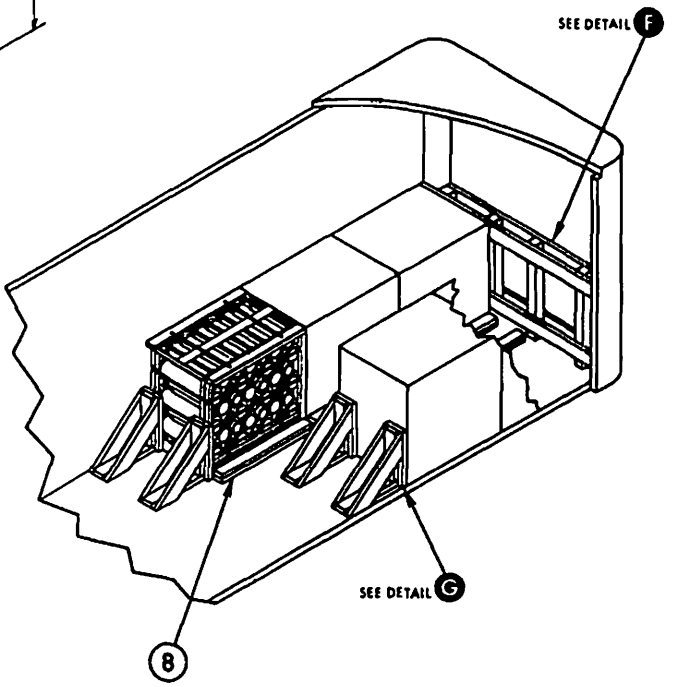
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NOTE

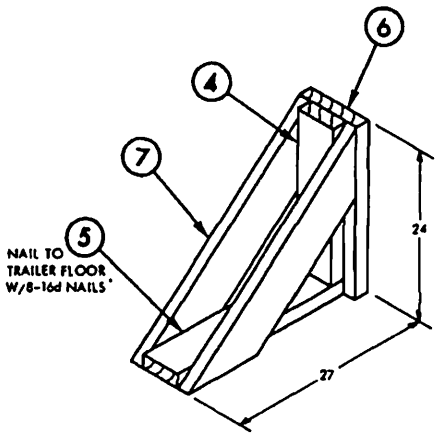
DO NOT USE TRAILERS WITH NON-NAILABLE FLOORS.



DETAIL F
FRONT BULKHEAD
1 REQUIRED



2 x 4 DOUBLED. NAIL 1ST PIECE TO TRAILER FLOOR W/1-16d NAIL EVERY 8 INCHES. NAIL 2ND PIECE TO 1ST IN LIKE MANNER.



DETAIL G
KNEE BRACE
QUANITY AS REQUIRED

*TW - TRAILER WIDTH

PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NAILS	
					NUMBER	SIZE
8	SLEEPER	2 x 4 x 45	10	SEE FIELD NOTE		
7	BRACE DIAGONAL	2 x 6 TO SUIT	8	4 & 5	4 PER JOINT	12d
6	BRACE BACKUP	2 x 10 x 24	4	4	5	10d
5	BRACE CLEAT	2 x 6 x 25 1/2	4	4	3	16d
4	BRACE UPRIGHT	2 x 6 x 22 1/2	4	SEE 5 & 6	-	-
3	FWD HORIZONTAL	2 x 6 TO SUIT	2	2	3 PER JOINT	12d
2	VERTICAL	4 x 4 x 43 3/4	4	SEE 1 & 2	-	-
1	AFT HORIZONTAL	2 x 6 x TW*-1	2	2	3 PER JOINT	12d

LIST OF MATERIALS AND NAILING DATA

TYPICAL LTL

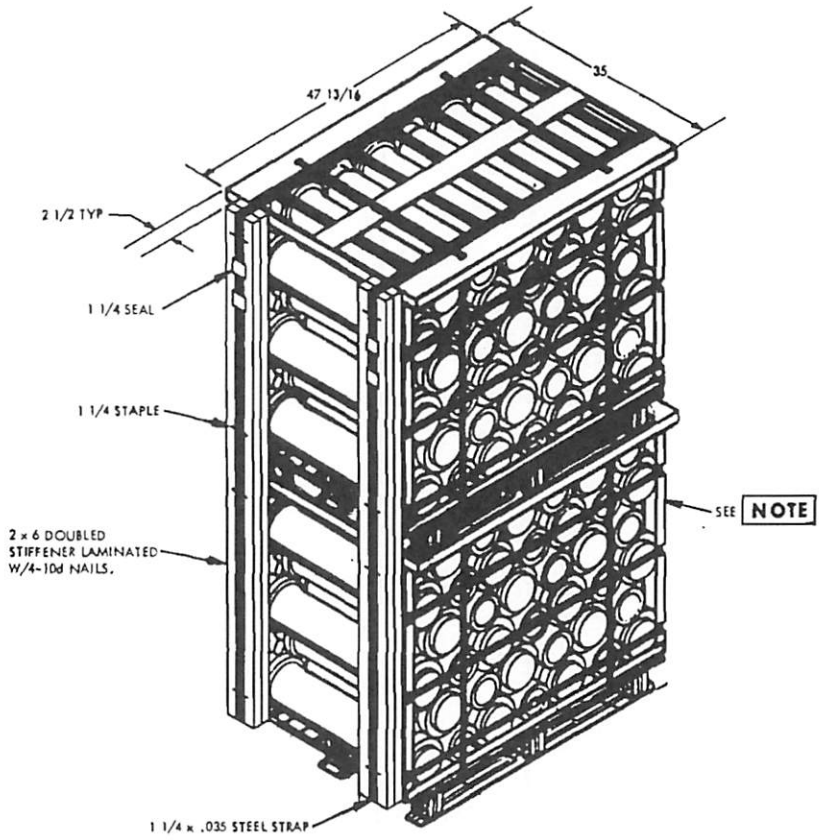
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UNITIZING PROCEDURE

1. WHEN REQUIRED BY THE FTL OR LTL REQUIREMENTS OF THIS MIL-STD, THE TWO HIGH STACKED UNIT LOADS SHALL BE STRAPPED TOGETHER AS SHOWN.
2. THE 2 x 6 DOUBLED STIFFENER SHALL EXTEND FROM THE TOP OF STACKED UNIT LOADS TO THE TOP OF THE PALLET OF THE BOTTOM UNIT LOAD. THE STIFFENER SHALL BE LAMINATED WITH FOUR 10d NAILS TO PREVENT SHIFTING.
3. THE 1 1/4 INCH STEEL STRAPS POSITION AS SHOWN ENCIRCLE THE STACKED UNIT LOADS AND PASS UNDER THE DECK OF THE BOTTOM PALLET WITH THE STRAPS HOLDING THE STIFFENERS IN PLACE.
4. TENSION STRAPS AND SEAL WITH TWO DOUBLED CRIMPED SEALS. SECURE EACH STRAP TO THE STIFFENER WITH FOUR 1 1/4 INCH STAPLES.

NOTE

WHEN LOADING A TRAILER WITH A SINGLE STACKED UNIT LOAD (NOT BACKED UP BY ANOTHER STACKED LOAD), STIFFENERS SHALL BE USED ON BOTH ENDS OF THE UNITIZED LOADS. SEE GENERAL NOTE 4.



REVIEW ACTIVITIES:
NAVY-OS, AS

PREPARING ACTIVITY:
NAVY - OS
(PROJECT NO. 8140-N-334)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE: MIL-STD-1320-84A (Navy)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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